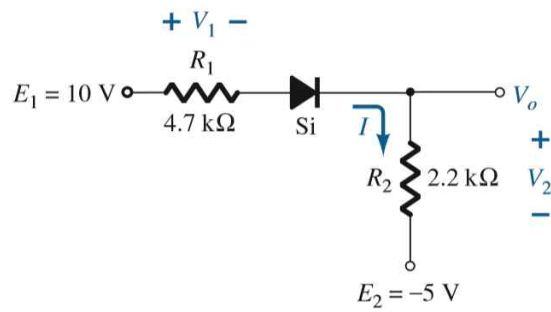


[1] Example 2.9

Determine I , V_1 , V_2 and V_o .



(solution)

$$I = \frac{E_1 - E_2 - V_D}{R_1 + R_2} = 2.07 \text{ mA}$$

$$V_1 = 9.73 \text{ V} , \quad V_2 = 4.55 \text{ V} ,$$

$$V_2 = V_o - E_2 \quad \therefore V_o = V_2 + E_2 = -0.45 \text{ V}$$

[2] (Example 2.13)

Determine I_1 , I_2 and I_{D_2} .

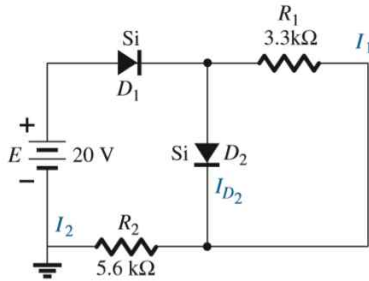


FIG. 37

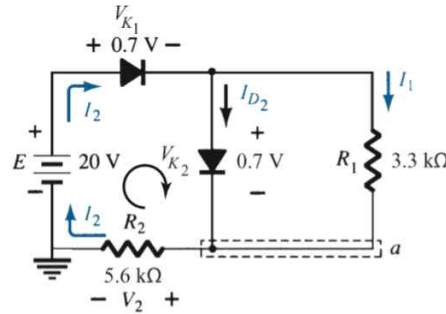


FIG. 38

Solution: The applied voltage (pressure) is such as to turn both diodes on, as indicated by the resulting current directions in the network of Fig. 38. Note the use of the abbreviated notation for “on” diodes and that the solution is obtained through an application of techniques applied to dc series-parallel networks. We have

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = \mathbf{0.212 \text{ mA}}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

and
$$V_2 = E - V_{K_1} - V_{K_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = \mathbf{18.6 \text{ V}}$$

with
$$I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = \mathbf{3.32 \text{ mA}}$$

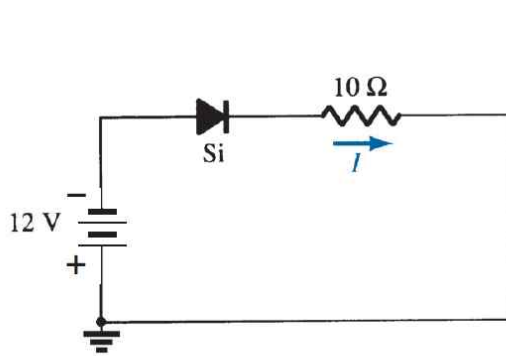
At the bottom node a ,

$$I_{D_2} + I_1 = I_2$$

and
$$I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} \cong \mathbf{3.11 \text{ mA}}$$

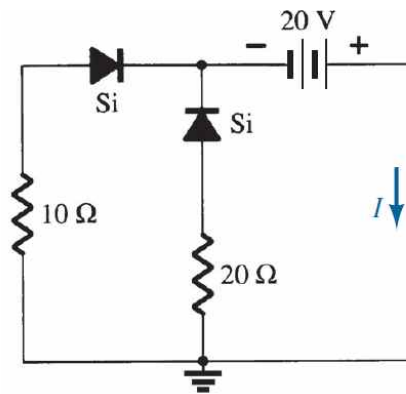
[3] Chap.2 Problem 5 (p.123)

5. Determine the current I for each of the configurations of Fig. 155 using the approximate equivalent model for the diode.



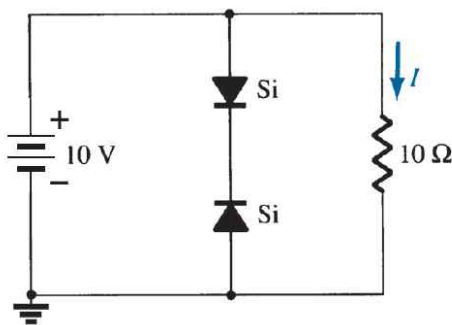
(a)

- (a) $I = 0 \text{ mA}$; diode reverse-biased.



(b)

$$\begin{aligned}
 (b) \quad V_{20\Omega} &= 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V} \\
 I(20 \Omega) &= \frac{19.3 \text{ V}}{20 \Omega} = 0.965 \text{ A} \\
 V(10 \Omega) &= 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V} \\
 I(10 \Omega) &= \frac{19.3 \text{ V}}{10 \Omega} = 1.93 \text{ A} \\
 I &= I(10 \Omega) + I(20 \Omega) \\
 &= 2.895 \text{ A}
 \end{aligned}$$

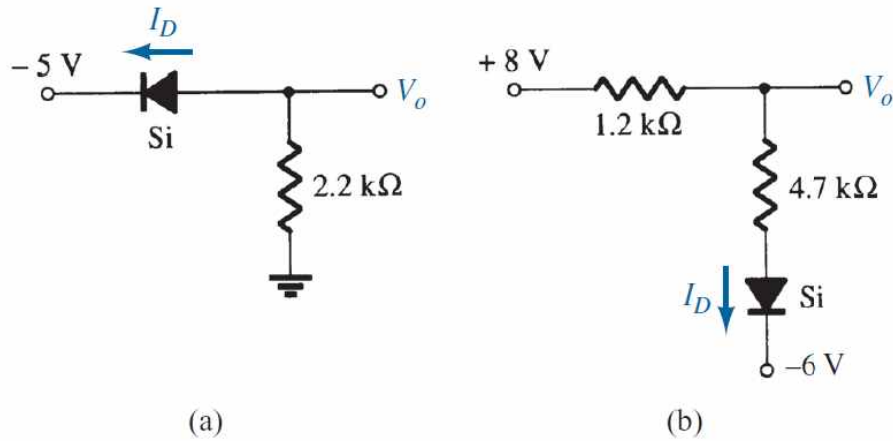


(c)

- (c) $I = \frac{10 \text{ V}}{10 \Omega} = 1 \text{ A}$; center branch open

[4] Chap.2 Problem 6 (p.123)

6. Determine V_o and I_D for the networks of Fig. 156.



6. (a) Diode forward-biased,
 Kirchhoff's voltage law (CW): $-5\text{ V} + 0.7\text{ V} - V_o = 0$
 $V_o = -4.3\text{ V}$

$$I_R = I_D = \frac{|V_o|}{R} = \frac{4.3\text{ V}}{2.2\text{ k}\Omega} = \mathbf{1.955\text{ mA}}$$

- (b) Diode forward-biased,

$$I_D = \frac{8\text{ V} + 6\text{ V} - 0.7\text{ V}}{1.2\text{ k}\Omega + 4.7\text{ k}\Omega} = \mathbf{2.25\text{ mA}}$$

$$V_o = 8\text{ V} - (2.25\text{ mA})(1.2\text{ k}\Omega) = \mathbf{5.3\text{ V}}$$

[5] Chap.2 Problem 8 (p.123)

*8. Determine V_o and I_D for the networks of Fig. 158.

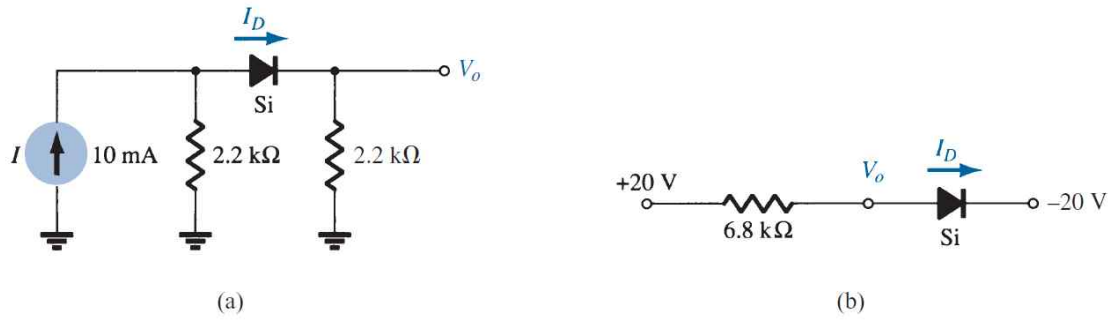


FIG. 158
Problem 8.

8. (a) $V_o = 10.65 \text{ V}, \quad I_D = 4.84 \text{ mA}$

(b) $V_o = -19.3 \text{ V}, \quad I_D = 5.78 \text{ mA}$

[6] Chap.2 Problem 12 (p.124)

12. Determine V_{o1} , V_{o2} , and I for the network of Fig. 162.

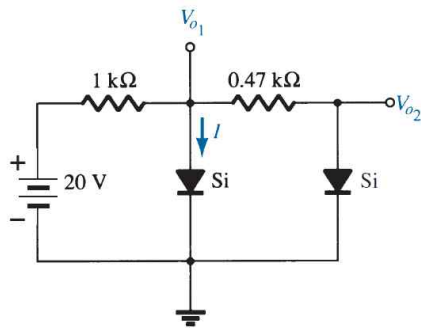


FIG. 162
Problem 12.

12. Both diodes forward-biased:

$$V_{o1} = 0.7 \text{ V}, V_{o2} = 0.7 \text{ V}$$

$$I_{1 \text{ k}\Omega} = \frac{20 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = \frac{19.3 \text{ V}}{1 \text{ k}\Omega} = 19.3 \text{ mA}$$

$$I_{0.47 \text{ k}\Omega} = 0 \text{ mA}$$

$$I = I_{1 \text{ k}\Omega} - I_{0.47 \text{ k}\Omega} = 19.3 \text{ mA} - 0 \text{ mA} \\ = \mathbf{19.3 \text{ mA}}$$

[7] Chap.2 Problem 14, 15, 16, 17, 20 (p.125)

14. Determine V_o for the network of Fig. 39 with 0 V on both inputs.
 15. Determine V_o for the network of Fig. 39 with 10 V on both inputs.
 16. Determine V_o for the network of Fig. 42 with 0 V on both inputs.
 17. Determine V_o for the network of Fig. 42 with 10 V on both inputs.

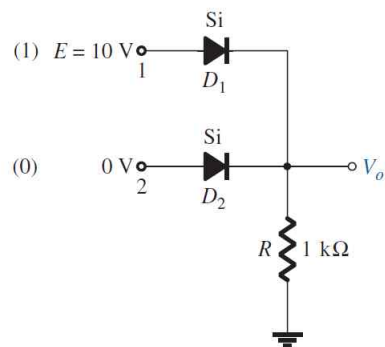


FIG. 39

Positive logic OR gate.

V_1	V_2	V_o
0 V	0 V	
0 V	10 V	
10 V	0 V	
10 V	10 V	

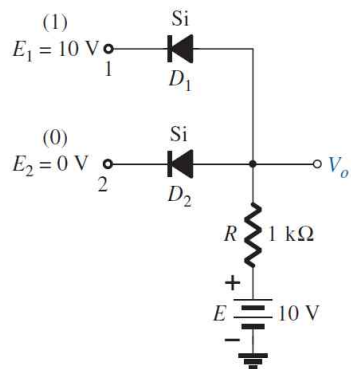


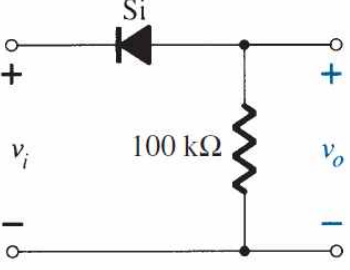
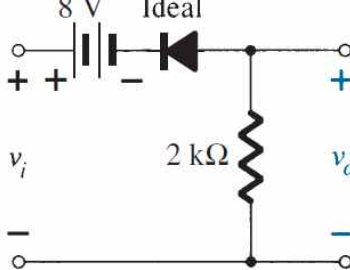
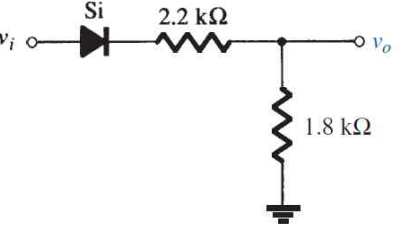
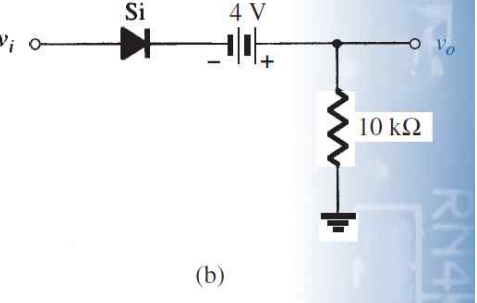
FIG. 42

Positive logic AND gate.

V_1	V_2	V_o
0 V	0 V	
0 V	10 V	
10 V	0 V	
10 V	10 V	

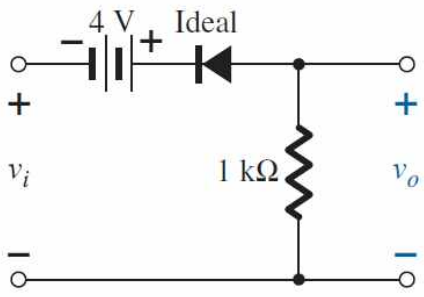
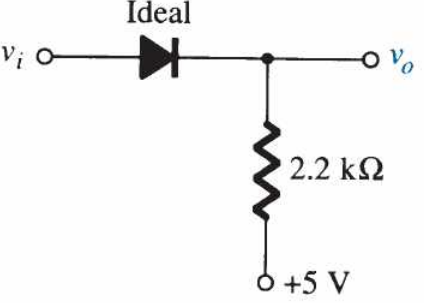
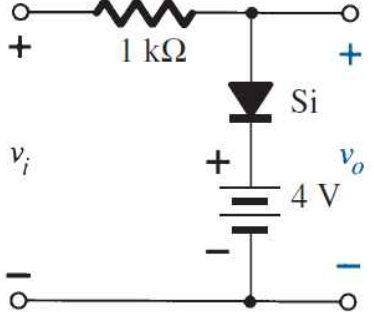
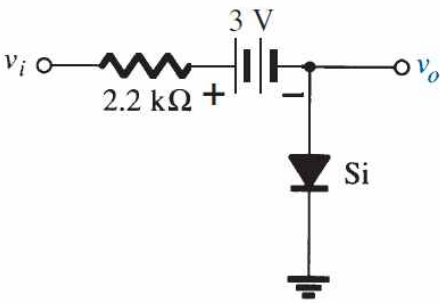
[8] Chap.2 Problem 32, 33 (p.127)

Determine v_o , when $v_i = 10\text{V}$, 5V , 0V , -5V and -10V .

	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													
	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													
 (a)	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													
 (b)	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													

[9] Chap.2 Problem 34, 35 (p.128)

Determine v_o , when $v_i = 10\text{V}, 5\text{V}, 0\text{V}, -5\text{V}$ and -10V .

	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													
	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													
	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													
	<table border="1"> <thead> <tr> <th>V_i</th><th>V_o</th></tr> </thead> <tbody> <tr><td>10 V</td><td></td></tr> <tr><td>5 V</td><td></td></tr> <tr><td>0 V</td><td></td></tr> <tr><td>-5 V</td><td></td></tr> <tr><td>-10 V</td><td></td></tr> </tbody> </table>	V_i	V_o	10 V		5 V		0 V		-5 V		-10 V	
V_i	V_o												
10 V													
5 V													
0 V													
-5 V													
-10 V													

[10] Chap.2 Problem 36 (p.128)

Determine v_o , when $v_i = 10\text{V}$, 5V , 0V , -5V and -10V .

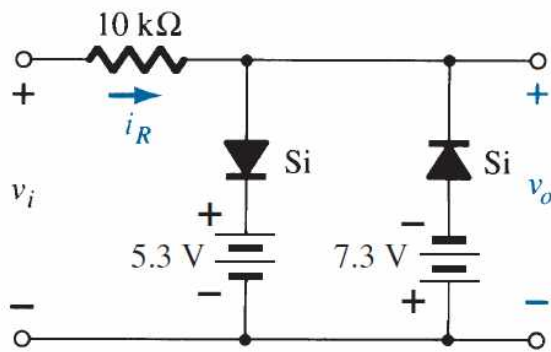
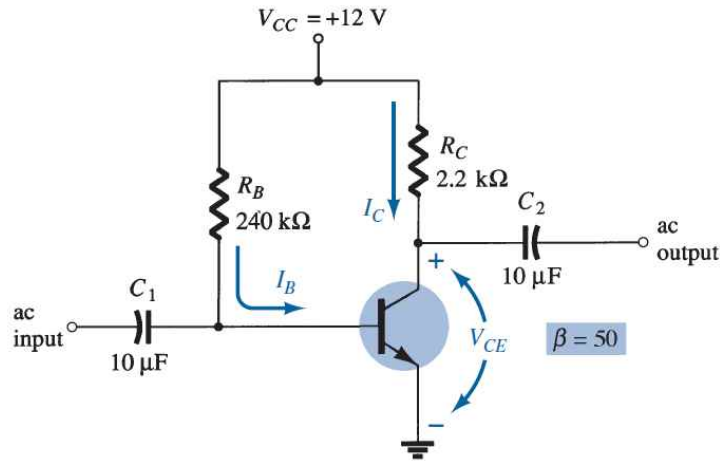


FIG. 180
Problem 36.

V_i	V_o
10 V	
5 V	
0 V	
-5 V	
-10 V	

[11] (Example 4.1)

Determine I_{B_Q} , I_{C_Q} , V_{CE_Q} , V_B , V_C , V_{BC} .

**FIG. 7**

DC fixed-bias circuit for Example 1.

- (1) $I_{B_Q} = \frac{12 - V_{BE}}{240k} = \frac{12 - 0.7}{240k} = 47.08 \mu A$
- (2) $I_{C_Q} = \beta I_{B_Q} = 50 \times 47.08 \mu A = 2.35 \text{ mA}$
- (3) $V_{CE_Q} = V_{CC} - I_C R_C = 12 - (2.35 \text{ mA})(2.2k\Omega) = 6.83 \text{ V}$
- (4) $V_B = V_{BE} = 0.7 \text{ V}$
- (5) $V_C = V_{CE} = 6.83 \text{ V}$
- (6) $V_{BC} = V_B - V_C = 0.7 - 6.83 = -6.13 \text{ V}$

[12] Chap.4 Problem 1 (p.244)

Determine

- a. I_{BQ} .
- b. I_{CQ} .
- c. V_{CEQ} .
- d. V_C .
- e. V_B .
- f. V_E .
- g. $I_{C_{sat}}$

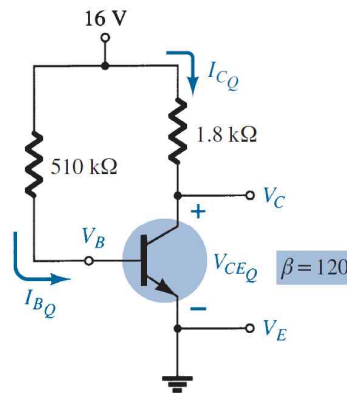


FIG. 118

1.
 - (a) $I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16 \text{ V} - 0.7 \text{ V}}{510 \text{ k}\Omega} = \frac{15.3 \text{ V}}{510 \text{ k}\Omega} = \mathbf{30 \mu A}$
 - (b) $I_{CQ} = \beta I_{BQ} = (120)(30 \mu A) = \mathbf{3.6 \text{ mA}}$
 - (c) $V_{CEQ} = V_{CC} - I_{CQ} R_C = 16 \text{ V} - (3.6 \text{ mA})(1.8 \text{ k}\Omega) = \mathbf{9.52 \text{ V}}$
 - (d) $V_C = V_{CEQ} = \mathbf{6.48 \text{ V}}$
 - (e) $V_B = V_{BE} = \mathbf{0.7 \text{ V}}$
 - (f) $V_E = \mathbf{0 \text{ V}}$

[13] Chap. 4 Problem 2 (p.244)

Determine

- a. I_C .
- b. R_C .
- c. R_B .
- d. V_{CE} .

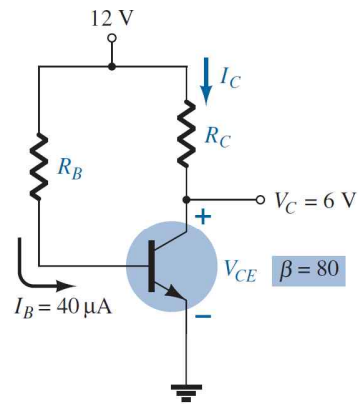


FIG. 119

2. (a) $I_C = \beta I_B = 80(40 \mu\text{A}) = \mathbf{3.2 \text{ mA}}$
- (b) $R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{12 \text{ V} - 6 \text{ V}}{3.2 \text{ mA}} = \frac{6 \text{ V}}{3.2 \text{ mA}} = \mathbf{1.875 \text{ k}\Omega}$
- (c) $R_B = \frac{V_{R_B}}{I_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{40 \mu\text{A}} = \frac{11.3 \text{ V}}{40 \mu\text{A}} = \mathbf{282.5 \text{ k}\Omega}$
- (d) $V_{CE} = V_C = \mathbf{6 \text{ V}}$

[14] Chap. 4 Problem 3 (p.244)

Determine

- a. I_C .
- b. V_{CC} .
- c. β .
- d. R_B .

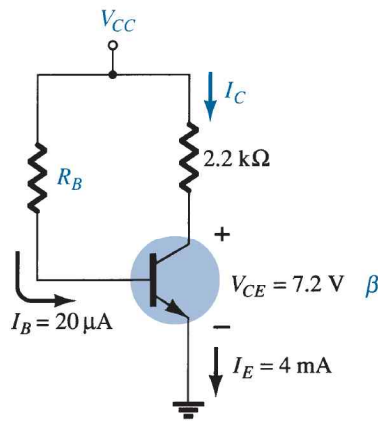


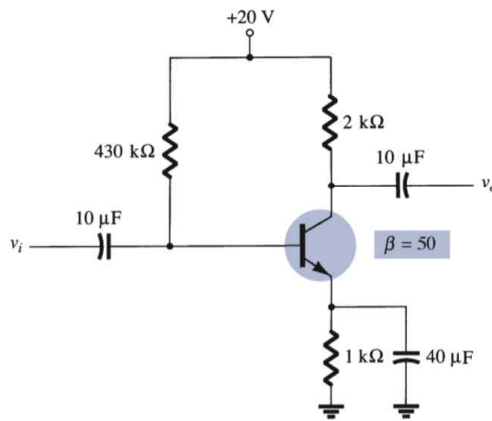
FIG. 120

- 3.
- (a) $I_C = I_E - I_B = 4 \text{ mA} - 20 \mu\text{A} = \mathbf{3.98 \text{ mA}} \cong 4 \text{ mA}$
 - (b) $V_{CC} = V_{CE} + I_C R_C = 7.2 \text{ V} + (3.98 \text{ mA})(2.2 \text{ k}\Omega)$
 $= \mathbf{15.96 \text{ V}} \cong 16 \text{ V}$
 - (c) $\beta = \frac{I_C}{I_B} = \frac{3.98 \text{ mA}}{20 \mu\text{A}} = \mathbf{199} \cong 200$
 - (d) $R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{15.96 \text{ V} - 0.7 \text{ V}}{20 \mu\text{A}} = \mathbf{763 \text{ k}\Omega}$

[15] (Example 4.4)

Determine

- I_B .
- I_C .
- V_{CE} .
- V_C .
- V_E .
- V_B .
- V_{BC} .

**Solution:**

- Eq. (17):
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{430 \text{ k}\Omega + (51)(1 \text{ k}\Omega)}$$
$$= \frac{19.3 \text{ V}}{481 \text{ k}\Omega} = \mathbf{40.1 \mu A}$$
- $I_C = \beta I_B$
$$= (50)(40.1 \mu A)$$
$$\cong \mathbf{2.01 \text{ mA}}$$
- Eq. (19):
$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$
$$= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega + 1 \text{ k}\Omega) = 20 \text{ V} - 6.03 \text{ V}$$
$$= \mathbf{13.97 \text{ V}}$$
- $V_C = V_{CC} - I_C R_C$
$$= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega) = 20 \text{ V} - 4.02 \text{ V}$$
$$= \mathbf{15.98 \text{ V}}$$
- $V_E = V_C - V_{CE}$
$$= 15.98 \text{ V} - 13.97 \text{ V}$$
$$= \mathbf{2.01 \text{ V}}$$

or $V_E = I_E R_E \cong I_C R_E$
$$= (2.01 \text{ mA})(1 \text{ k}\Omega)$$
$$= \mathbf{2.01 \text{ V}}$$
- $V_B = V_{BE} + V_E$
$$= 0.7 \text{ V} + 2.01 \text{ V}$$
$$= \mathbf{2.71 \text{ V}}$$
- $V_{BC} = V_B - V_C$
$$= 2.71 \text{ V} - 15.98 \text{ V}$$
$$= \mathbf{-13.27 \text{ V}} \text{ (reverse-biased as required)}$$

[16] Chap. 4 Problem 8 (p.245)

Determine

- a. I_{BQ} .
- b. I_{CQ} .
- c. V_{CEQ} .
- d. V_C .
- e. V_B .
- f. V_E .
- g. $I_{C\text{sat}}$

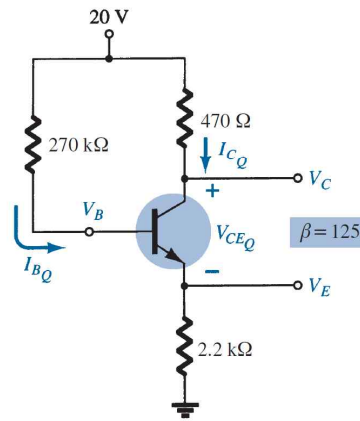


FIG. 122

8. (a)
$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\text{ V} - 0.7\text{ V}}{270\text{ k}\Omega + (126)2.2\text{ k}\Omega} = \frac{19.3\text{ V}}{547.2\text{ k}\Omega} = \mathbf{35.27\text{ }\mu\text{A}}$$
- (b)
$$I_{CQ} = \beta I_{BQ} = (125)(35.27\text{ }\mu\text{A}) = \mathbf{4.41\text{ mA}}$$
- (c)
$$\begin{aligned} V_{CEQ} &= V_{CC} - I_C(R_C + R_E) = 20\text{ V} - (4.41\text{ mA})(470\text{ k}\Omega + 2.2\text{ k}\Omega) \\ &= 20\text{ V} - 11.77\text{ V} \\ &= \mathbf{8.23\text{ V}} \end{aligned}$$
- (d)
$$\begin{aligned} V_C &= V_{CC} - I_C R_C = 20\text{ V} - (4.41\text{ mA})(470\text{ k}\Omega) = 20\text{ V} - 2.07\text{ V} \\ &= \mathbf{17.93\text{ V}} \end{aligned}$$
- (e)
$$\begin{aligned} V_B &= V_{CC} - I_B R_B = 20\text{ V} - (35.27\text{ }\mu\text{A})(270\text{ k}\Omega) \\ &= 20\text{ V} - 9.52\text{ V} = \mathbf{10.48\text{ V}} \end{aligned}$$
- (f)
$$V_E = V_C - V_{CE} = 17.93\text{ V} - 8.23\text{ V} = \mathbf{9.7\text{ V}}$$

[17] Chap. 4 Problem 10 (p.246)

Determine

- a. R_C .
- b. R_E .
- c. R_B .
- d. V_{CE} .
- e. V_B .

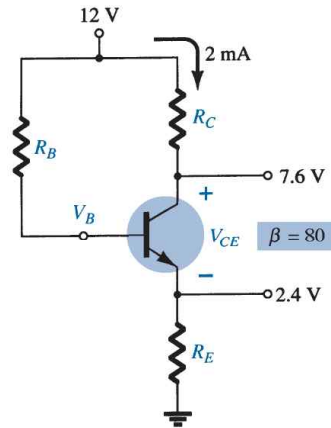


FIG. 123

10. (a) $R_C = \frac{V_{CC} - V_C}{I_C} = \frac{12\text{ V} - 7.6\text{ V}}{2\text{ mA}} = \frac{4.4\text{ V}}{2\text{ mA}} = \mathbf{2.2\text{ k}\Omega}$
- (b) $I_E \cong I_C: R_E = \frac{V_E}{I_E} = \frac{2.4\text{ V}}{2\text{ mA}} = \mathbf{1.2\text{ k}\Omega}$
- (c) $R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{12\text{ V} - 0.7\text{ V} - 2.4\text{ V}}{2\text{ mA}/80} = \frac{8.9\text{ V}}{25\text{ }\mu\text{A}} = \mathbf{356\text{ k}\Omega}$
- (d) $V_{CE} = V_C - V_E = 7.6\text{ V} - 2.4\text{ V} = \mathbf{5.2\text{ V}}$
- (e) $V_B = V_{BE} + V_E = 0.7\text{ V} + 2.4\text{ V} = \mathbf{3.1\text{ V}}$

[18] Chap. 4 Problem 11 (p.246)

Determine

- a. β .
- b. V_{CC} .
- c. R_B .

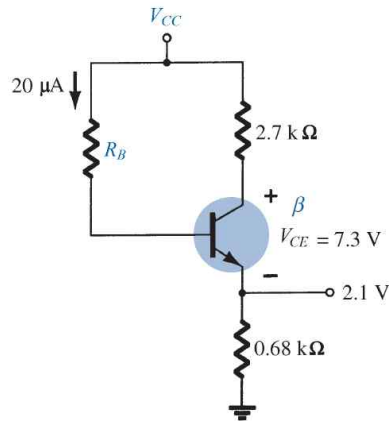
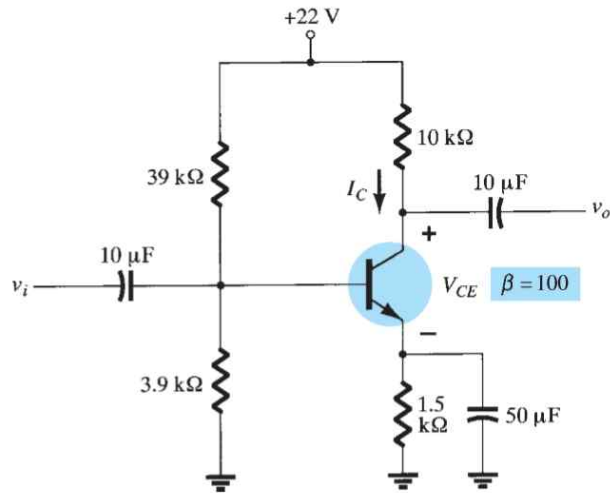


FIG. 124

11. (a) $I_C \cong I_E = \frac{V_E}{R_E} = \frac{2.1 \text{ V}}{0.68 \text{ k}\Omega} = 3.09 \text{ mA}$
- $$\beta = \frac{I_C}{I_B} = \frac{3.09 \text{ mA}}{20 \mu\text{A}} = \mathbf{154.5}$$
- (b) $V_{CC} = V_{R_C} + V_{CE} + V_E$
- $$= (3.09 \text{ mA})(2.7 \text{ k}\Omega) + 7.3 \text{ V} + 2.1 \text{ V} = 8.34 \text{ V} + 7.3 \text{ V} + 2.1 \text{ V}$$
- $$= \mathbf{17.74 \text{ V}}$$
- (c) $R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{17.74 \text{ V} - 0.7 \text{ V} - 2.1 \text{ V}}{20 \mu\text{A}}$
- $$= \frac{14.94 \text{ V}}{20 \mu\text{A}} = \mathbf{747 \text{ k}\Omega}$$

[19] (Example 4.8)

Determine the V_{CE} and I_C .



$$\begin{aligned} \text{Eq. (4.28): } R_{Th} &= R_1 \parallel R_2 \\ &= \frac{(39 \text{ k}\Omega)(3.9 \text{ k}\Omega)}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 3.55 \text{ k}\Omega \end{aligned}$$

$$\begin{aligned} \text{Eq. (4.29): } E_{Th} &= \frac{R_2 V_{CC}}{R_1 + R_2} \\ &= \frac{(3.9 \text{ k}\Omega)(22 \text{ V})}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} = 2 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{Eq. (4.30): } I_B &= \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \\ &= \frac{2 \text{ V} - 0.7 \text{ V}}{3.55 \text{ k}\Omega + (101)(1.5 \text{ k}\Omega)} = \\ &= 8.38 \mu\text{A} \end{aligned}$$

$$\begin{aligned} I_C &= \beta I_B \\ &= (100)(8.38 \mu\text{A}) \\ &= \mathbf{0.84 \text{ mA}} \end{aligned}$$

$$\begin{aligned} \text{Eq. (4.31): } V_{CE} &= V_{CC} - I_C(R_C + R_E) \\ &= 22 \text{ V} - (0.84 \text{ mA})(10 \text{ k}\Omega + 1.5 \text{ k}\Omega) \\ &= 22 \text{ V} - 9.66 \text{ V} \\ &= \mathbf{12.34 \text{ V}} \end{aligned}$$

[20] Chap. 4 Problem 15 (p.247)

Determine

- a. I_{BQ} .
- b. I_{CQ} .
- c. V_{CEQ} .
- d. V_C .
- e. V_E .
- f. V_B .

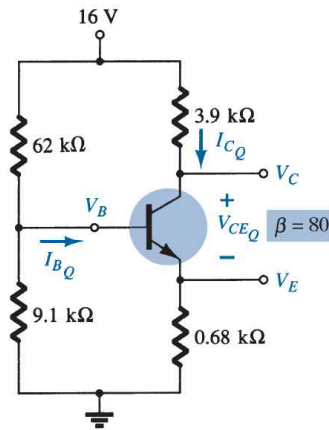


FIG. 125

15. $\beta R_E \geq 10R_2$
 $(80)(0.68 \text{ k}\Omega) \geq 10(9.1 \text{ k}\Omega)$
 $54.4 \text{ k}\Omega \not\geq 91 \text{ k}\Omega$ (No!)
- (a) Use exact approach:
 $R_{Th} = R_1 \parallel R_2 = 62 \text{ k}\Omega \parallel 9.1 \text{ k}\Omega = 7.94 \text{ k}\Omega$
 $E_{Th} = \frac{R_2 V_{CC}}{R_2 + R_1} = \frac{(9.1 \text{ k}\Omega)(16 \text{ V})}{9.1 \text{ k}\Omega + 62 \text{ k}\Omega} = 2.05 \text{ V}$
 $I_{BQ} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.05 \text{ V} - 0.7 \text{ V}}{7.94 \text{ k}\Omega + (81)(0.68 \text{ k}\Omega)}$
 $= \mathbf{21.42 \mu A}$
- (b) $I_{CQ} = \beta I_{BQ} = (80)(21.42 \mu A) = \mathbf{1.71 \text{ mA}}$
- (c) $V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E)$
 $= 16 \text{ V} - (1.71 \text{ mA})(3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega)$
 $= \mathbf{8.17 \text{ V}}$
- (d) $V_C = V_{CC} - I_C R_C$
 $= 16 \text{ V} - (1.71 \text{ mA})(3.9 \text{ k}\Omega)$
 $= \mathbf{9.33 \text{ V}}$
- (e) $V_E = I_E R_E \cong I_C R_E = (1.71 \text{ mA})(0.68 \text{ k}\Omega)$
 $= \mathbf{1.16 \text{ V}}$
- (f) $V_B = V_E + V_{BE} = 1.16 \text{ V} + 0.7 \text{ V}$
 $= \mathbf{1.86 \text{ V}}$

[21] Chap. 4 Problem 17 (p.247)

Determine

- a. I_C .
- b. V_E .
- c. V_B .
- d. R_1 .
- e. $I_{C\text{sat}}$

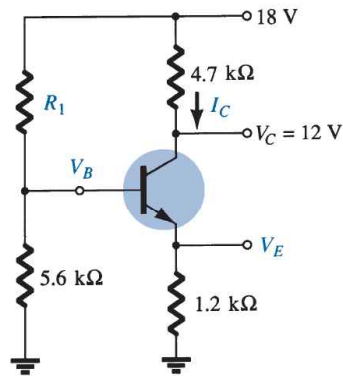


FIG. 126

$$17. \quad (a) \quad I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18 \text{ V} - 12 \text{ V}}{4.7 \text{ k}\Omega} = \mathbf{1.28 \text{ mA}}$$

$$(b) \quad V_E = I_E R_E \cong I_C R_E = (1.28 \text{ mA})(1.2 \text{ k}\Omega) = \mathbf{1.54 \text{ V}}$$

$$(c) \quad V_B = V_{BE} + V_E = 0.7 \text{ V} + 1.54 \text{ V} = \mathbf{2.24 \text{ V}}$$

$$(d) \quad R_1 = \frac{V_{R_1}}{I_{R_1}}; \quad V_{R_1} = V_{CC} - V_B = 18 \text{ V} - 2.24 \text{ V} = \mathbf{15.76 \text{ V}}$$

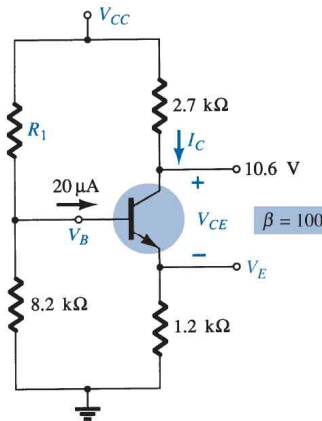
$$I_{R_1} \cong I_{R_2} = \frac{V_B}{R_2} = \frac{2.24 \text{ V}}{5.6 \text{ k}\Omega} = 0.4 \text{ mA}$$

$$R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{15.76 \text{ V}}{0.4 \text{ mA}} = \mathbf{39.4 \text{ k}\Omega}$$

[22] Chap. 4 Problem 18 (p.247)

Determine

- a. I_C .
- b. V_E .
- c. V_{CC} .
- d. V_{CE} .
- e. V_B .
- f. R_1 .

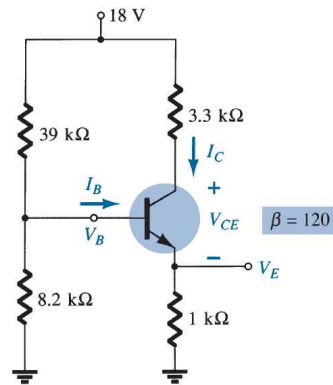
**FIG. 127**

18. (a) $I_C = \beta I_B = (100)(20 \mu\text{A}) = \mathbf{2 \text{ mA}}$
- (b) $I_E = I_C + I_B = 2 \text{ mA} + 20 \mu\text{A}$
 $= 2.02 \text{ mA}$
 $V_E = I_E R_E = (2.02 \text{ mA})(1.2 \text{ k}\Omega)$
 $= \mathbf{2.42 \text{ V}}$
- (c) $V_{CC} = V_C + I_C R_C = 10.6 \text{ V} + (2 \text{ mA})(2.7 \text{ k}\Omega)$
 $= 10.6 \text{ V} + 5.4 \text{ V}$
 $= \mathbf{16 \text{ V}}$
- (d) $V_{CE} = V_C - V_E = 10.6 \text{ V} - 2.42 \text{ V}$
 $= \mathbf{8.18 \text{ V}}$
- (e) $V_B = V_E + V_{BE} = 2.42 \text{ V} + 0.7 \text{ V} = \mathbf{3.12 \text{ V}}$
- (f) $I_{R_1} = I_{R_2} + I_B$
 $= \frac{3.12 \text{ V}}{8.2 \text{ k}\Omega} + 20 \mu\text{A} = 380.5 \mu\text{A} + 20 \mu\text{A} = 400.5 \mu\text{A}$
 $R_1 = \frac{V_{CC} - V_B}{I_{R_1}} = \frac{16 \text{ V} - 3.12 \text{ V}}{400.5 \mu\text{A}} = \mathbf{32.16 \text{ k}\Omega}$

[23] Chap. 4 Problem 21 (p.248)

Determine the following for the voltage-divider configuration of Fig. 128 using the approximate approach if the condition established by Eq. (33) is satisfied.

- a. I_C .
- b. V_{CE} .
- c. I_B .
- d. V_E .
- e. V_B .

**FIG. 128**

21. (a) $\beta R_E \geq 10R_2$
 $(120)(1 \text{ k}\Omega) \geq 10(8.2 \text{ k}\Omega)$
 $120 \text{ k}\Omega \geq 82 \text{ k}\Omega$ (checks)
 $\therefore V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{(8.2 \text{ k}\Omega)(18 \text{ V})}{39 \text{ k}\Omega + 8.2 \text{ k}\Omega} = 3.13 \text{ V}$
 $V_E = V_B - V_{BE} = 3.13 \text{ V} - 0.7 \text{ V} = 2.43 \text{ V}$
 $I_C \cong I_E = \frac{V_E}{R_E} = \frac{2.43 \text{ V}}{1 \text{ k}\Omega} = \mathbf{2.43 \text{ mA}}$
- (b) $V_{CE} = V_{CC} - I_C(R_C + R_E)$
 $= 18 \text{ V} - (2.43 \text{ mA})(3.3 \text{ k}\Omega + 1 \text{ k}\Omega)$
 $= \mathbf{7.55 \text{ V}}$
- (c) $I_B = \frac{I_C}{\beta} = \frac{2.43 \text{ mA}}{120} = \mathbf{20.25 \mu\text{A}}$
- (d) $V_E = I_E R_E \cong I_C R_E = (2.43 \text{ mA})(1 \text{ k}\Omega) = \mathbf{2.43 \text{ V}}$
- (e) $V_B = \mathbf{3.13 \text{ V}}$